

LECTURE 04

LINES

Let l be a line that is not parallel to the y -axis, and let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be distinct points on l . The slope m of l is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

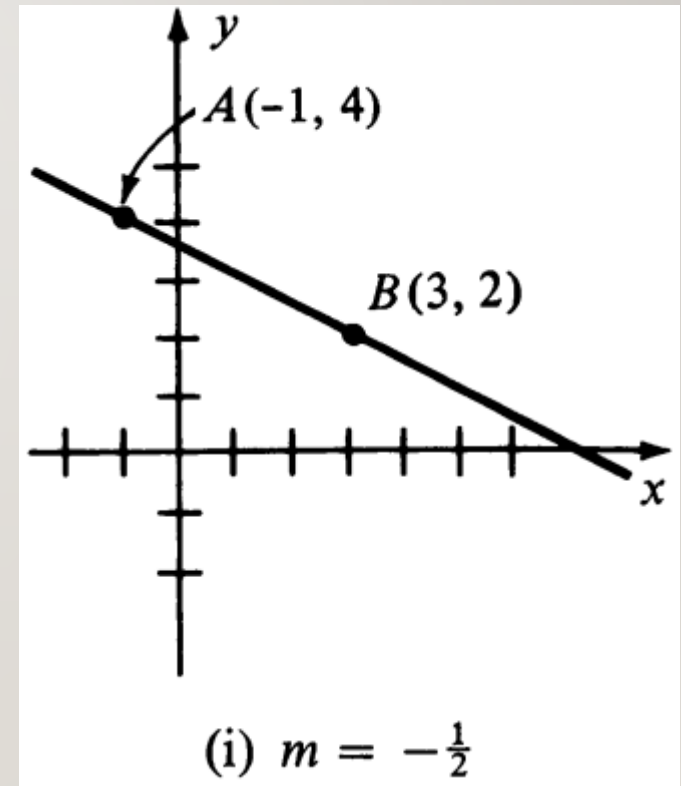
If l is parallel to the y -axis, then the slope is not defined.

EXAMPLE 01

Sketch the lines through the following pairs of points and find their slopes

(a) $A(-1, 4)$ and $B(3, 2)$

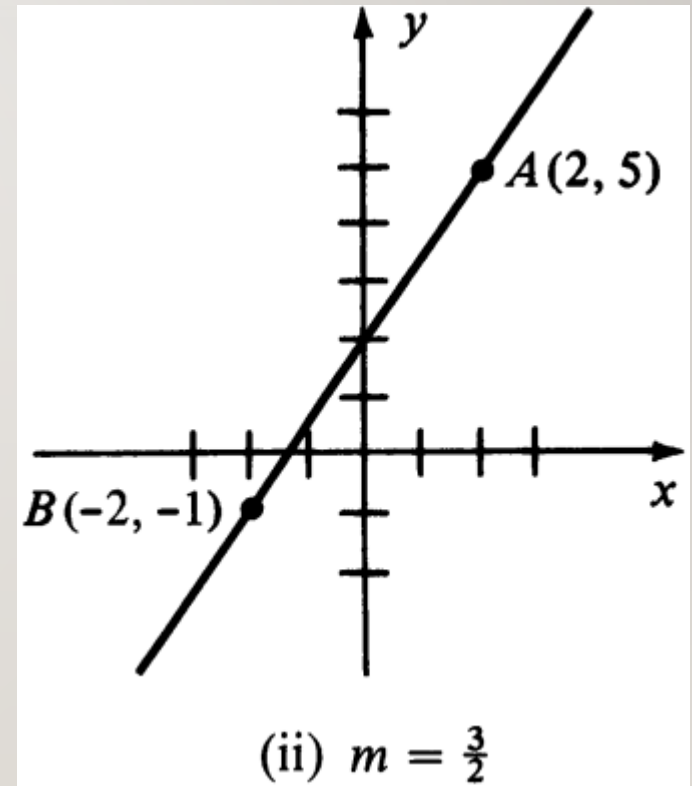
$$\text{Slope}(m) = \frac{2-4}{3-(-1)} = \frac{-2}{4} = -\frac{1}{2}$$



CONT...

(b) $A(2, 5)$ *and* $B(-2, -1)$

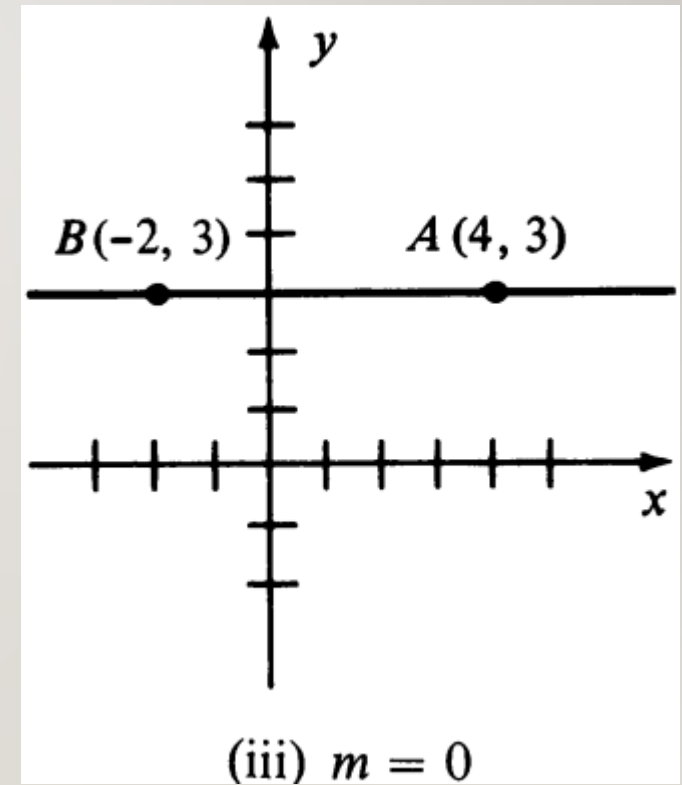
$$\text{Slope}(m) = \frac{-1-5}{-2-2} = \frac{-6}{-4} = \frac{3}{2}$$



CONT...

(c) $A(4, 3)$ and $B(-2, 3)$

$$\text{Slope}(m) = \frac{3-3}{-2-4} = \frac{0}{-6} = 0$$

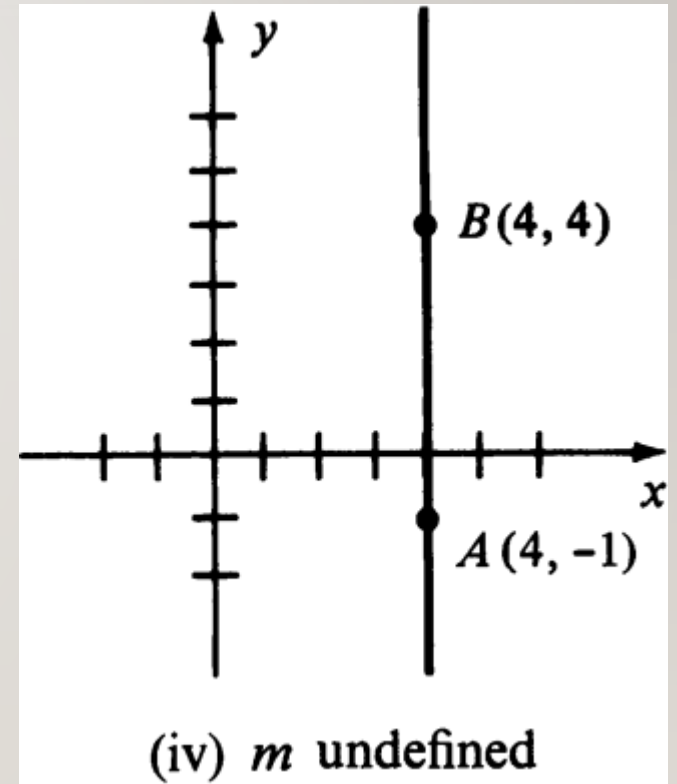


CONT...

(d) $A(4, -1)$ *and* $B(4, 4)$

$$\text{Slope}(m) = \frac{4 - (-1)}{4 - 4} = \frac{5}{0} = \textit{undefined}$$

The slope is undefined since the line is vertical.



POINT-SLOPE FORM

An equation for the line through the point $P(x_1, y_1)$ with slope m is

$$y - y_1 = m(x - x_1)$$

Example 02

Find an equation of the line through the points $A(1, 7)$ and $B(-3, 2)$.

Solution:

By the definition of slope m of the line is

$$m = \frac{7 - 2}{1 - (-3)} = \frac{5}{4}$$

CONT...

Using the coordinates of A in the point-slop form, we have

$$y - 7 = \frac{5}{4}(x - 1)$$

which is equivalent to

$$4y - 28 = 5x - 5 \quad \text{or} \quad 5x - 4y + 23 = 0.$$

The same equation would have been obtained if the coordinates of point B had been substituted.

SLOPE-INTERCEPT FORM

The graph of the equation $y = mx + b$ is a line having slope m and y -intercept b .

The graph of a linear equation $ax + by + c = 0$ is a line and, conversely, every line is the graph of a linear equation.

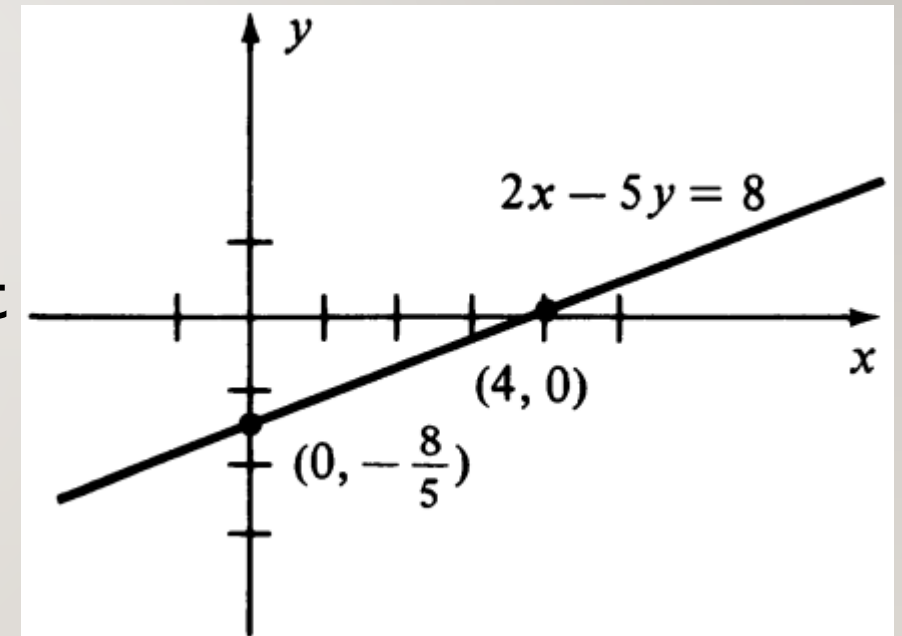
Example 03 Sketch the graph of $2x - 5y = 8$.

Solution: The graph is a line, and hence it is sufficient to find two points on the graph. Let us find the x -intercept and y -intercept.

CONT...

Substituting $y = 0$ in the given equation we obtain the x-intercept 4.

Substituting $x = 0$, we see that y-intercept is $-\frac{8}{5}$. This leads to the graph as shown in figure



PARALLEL LINES

Two nontrivial lines are parallel if and only if they have the same slope.

Example 04

Find an equation of a line through the point $(5, -7)$ that is parallel to line $6x + 3y - 4 = 0$.

Solution: Express the given equation in slope-intercept form. We begin by writing

$$3y = -6x + 4$$

CONT...

and then divide both sides by 3, obtaining

$$y = -2x + \frac{4}{3}.$$

The last equation is in slope-intercept form with $m=-2$ and hence the slope is -2. Since parallel lines have the same slope, the required line also has slope -2. Applying the point-slope form gives us

$$y + 7 = -2(x - 5).$$

This is equivalent to

$$y + 7 = -2x + 10 \quad \text{or} \quad 2x + y - 3 = 0$$

PERPENDICULAR LINES

Two lines with slope m_1 and m_2 are perpendicular if and only if $m_1 m_2 = -1$.

Example 05 Find an equation for the perpendicular bisector of the line segment from $A(1, 7)$ to $B(-3, 2)$.

Solution:

By the midpoint formula, the midpoint M of the segment AB is $(-1, 9/2)$. The slope of AB is

$$m = \frac{7 - 2}{1 - (-3)} = \frac{5}{4}$$

CONT...

It follows that the slope of the perpendicular bisector is $-4/5$.
Applying the point-slope form,

$$y - \frac{9}{2} = -\frac{4}{5}(x + 1).$$

Multiplying both sides by 10 and simplifying leads to
 $8x + 10y - 37 = 0.$

POLYGON

- A polygon is a closed figure determined by three or more line segments that lie in a plane.
- The line segment that form the polygon are called the sides.
- The name of the polygon is based on the number of its sides.

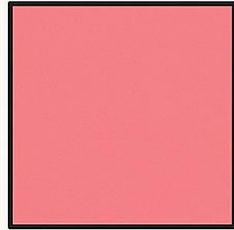
Number of sides	Name of Polygon
3	Triangle
4	Quadrilateral
5	Pentagon

QUADRILATERAL

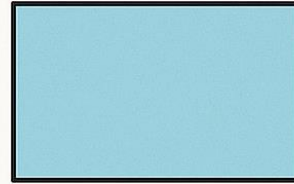
A **quadrilateral** is a polygon with **four sides (edges)** and **four vertices (corners)**.

- **Square** – all sides equal, all angles 90° .
- **Rectangle** – opposite sides equal, all angles 90° .
- **Parallelogram** – opposite sides parallel and equal, opposite angles equal.
- **Trapezium (Trapezoid)** – one pair of opposite sides parallel.

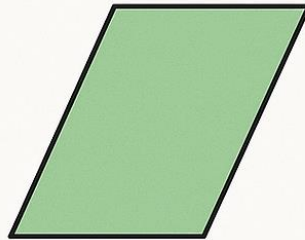
SHAPES



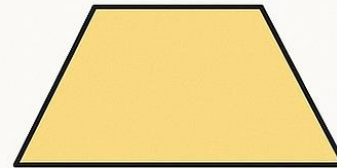
Square



Rectangle



Parallelogram



Trapezium

In Exercises 1–4 plot the points A and B and find the slope of the line through A and B .

1 $A(-4, 6), B(-1, 18)$

2 $A(6, -2), B(-3, 5)$

3 $A(-1, -3), B(-1, 2)$

4 $A(-3, 4), B(2, 4)$

5 Show that $A(-3, 1)$, $B(5, 3)$, $C(3, 0)$, and $D(-5, -2)$ are vertices of a parallelogram.

- 6 Show that $A(2, 3)$, $B(5, -1)$, $C(0, -6)$, and $D(-6, 2)$ are vertices of a trapezoid.
- 7 Prove that the points $A(6, 15)$, $B(11, 12)$, $C(-1, -8)$, and $D(-6, -5)$ are vertices of a rectangle.
- 8 Prove that the points $A(1, 4)$, $B(6, -4)$, and $C(-15, -6)$ are vertices of a right triangle.

In Exercises 11–24 find an equation for the line satisfying the given conditions.

11 Through $A(2, -6)$, slope $\frac{1}{2}$

12 Slope -3 , y -intercept 5

13 Through $A(-5, -7)$, $B(3, -4)$

14 x -intercept -4 , y -intercept 8

15 Through $A(8, -2)$, y -intercept -3

16 Slope 6 , x -intercept -2

17 Through $A(10, -6)$, parallel to (a) the y -axis; (b) the x -axis.

18 Through $A(-5, 1)$, perpendicular to (a) the y -axis; (b) the x -axis.

- 19** Through $A(7, -3)$, perpendicular to the line with equation $2x - 5y = 8$.
- 20** Through $(-\frac{3}{4}, -\frac{1}{2})$, parallel to the line with equation $x + 3y = 1$.
- 21** Given $A(3, -1)$ and $B(-2, 6)$, find an equation for the perpendicular bisector of the line segment AB .
- 22** Find an equation for the line which bisects the second and fourth quadrants.
- 23** Find equations for the altitudes of the triangle with vertices $A(-3, 2)$, $B(5, 4)$, $C(3, -8)$, and find the point at which they intersect.
- 24** Find equations for the medians of the triangle in Exercise 23, and find their point of intersection.

In Exercises 25–34 use the Slope-Intercept Form (1.17) to find the slope and y -intercept of the line with the given equation and sketch the graph of each line.

25 $3x - 4y + 8 = 0$

26 $2y - 5x = 1$

27 $x + 2y = 0$

28 $8x = 1 - 4y$

29 $y = 4$

30 $x + 2 = \frac{1}{2}y$

31 $5x + 4y = 20$

32 $y = 0$

33 $x = 3y + 7$

34 $x - y = 0$

35 Find a real number k such that the point $P(-1, 2)$ is on the line $kx + 2y - 7 = 0$.

36 Find a real number k such that the line $5x + ky - 3 = 0$ has y -intercept -5 .